TEZPUR UNIVERSITY

Spring Semester Mid-term Examination, 2018

CO205: Formal Languages & Automata (B.Tech)

Full Marks : 40 | Time : 75 mins

1. Give *one word* answers. [2]

   (a) The process of converting an NFA to a DFA is called ____________.

   (b) A string of variables and terminals that can be derived from the start variable in a Context Free Grammar is called ____________.

2. What is *ambiguity* in a Context Free Grammar? Give example. [3+3]

3. (a) State the Pumping lemma for regular languages. [3]

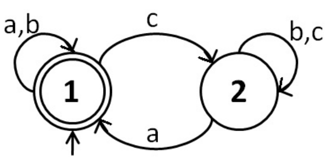
   (b) Prove that the language L is non-regular. [7]

$$L = \{w | w \in \Sigma^*, \Sigma = \{0, 1, +, =\} \text{ and } w \text{ is of the form } x = y + z \text{ where } x, y, z \text{ are binary integers and } x \text{ is the sum of } y \text{ and } z\}$$

4. Construct regular expressions for the following regular languages:

   (a) L = $\{w \in \{a, b, c\}^* | w$ has a substring with odd number of a's$\}$ [3]

   (b) L = language of the following DFA (Show the derivation steps) [8]



5. (a) What is ε-closure of a state? [3]

   (b) Minimize the following DFA (show the derivation steps). [8]